



VPC Traffic Flow and Security



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sg-0d6b299c7467bd615 - NextWork Security Group Actions ▾

Details

Security group name NextWork Security Group	Security group ID sg-0d6b299c7467bd615	Description A Security Group for the NextWork VPC	VPC ID vpc-0a4c28603f48a4789
Owner 011528261245	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

[Inbound rules](#) | [Outbound rules](#) | [Tags](#)

Inbound rules (1) Refresh Manage tags Edit inbound rules

Search

☐	Name ▾	Security group rule... ▾	IP version ▾	Type ▾	Protocol ▾	Port range
☐	-	sgr-09cf12eb87b726238	IPv4	HTTP	TCP	80



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) lets you create isolated cloud resources in a virtual network. It's useful because it offers control over IP ranges, subnets, routing, and security, allowing secure communication and management of cloud infra.

How I used Amazon VPC in this project

I used Amazon VPC in today's project to create isolated subnets for my EC2 instances, configured route tables for internet access, and applied security groups and network ACLs to control traffic flow securely.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was the need to adjust both security groups and network ACLs to fully enable traffic flow between subnets and the internet, requiring more granular configuration than anticipated.

This project took me...

It took me 1 hr to complete this project.



Route tables

Route tables are essential network components that define how traffic is directed across networks or subnets. They list routes with destination IPs, target gateways, and metrics, ensuring efficient data flow and connectivity within and between network

Route tables are needed to make a subnet public because they define how traffic is directed to and from the internet. A public subnet requires a route table entry that directs outbound traffic to an internet gateway, allowing resources to communicate

VPC > Route tables > rtb-048728810ba8b8eb0

rtb-048728810ba8b8eb0 / NextWork route table Actions ▼

Details Info

Route table ID rtb-048728810ba8b8eb0	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-07d1156cca87b2189 NextWork VPC	Owner ID 011528261245		

Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (2) Both ▼ Edit routes

Destination	Target	Status	Propagated
0.0.0.0/0	igw-02995947b38b9ba61	Active	No
10.0.0.0/16	local	Active	No



Route destination and target

Routes are defined by their destination and target, which means the destination specifies the IP address range for the traffic, while the target indicates where the traffic should be directed, such as a gateway, network interface, or another subnet.

The route in my route table that directed internet-bound traffic to my internet gateway had a destination of `0.0.0.0/0` and a target of my internet gateway's ID, allowing all outbound traffic to reach the internet through that gateway.

The screenshot shows the AWS Management Console interface for a route table. The breadcrumb navigation at the top reads: VPC > Route tables > rtb-048728810ba8b8eb0. The title of the page is "rtb-048728810ba8b8eb0 / NextWork route table".

Details Info

Route table ID rtb-048728810ba8b8eb0	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-07d1156cca87b2189 NextWork VPC	Owner ID 011528261245		

Routes Subnet associations Edge associations Route propagation Tags

Routes (2) Both Edit routes

Filter routes

Destination	Target	Status	Propagated
0.0.0.0/0	igw-02995947b38b9ba61	Active	No
10.0.0.0/16	local	Active	No



Security groups

Security groups are virtual firewalls in cloud environments that manage inbound and outbound traffic for resources like EC2 instances. They contain rules defining allowed protocols, ports, and IP address ranges to enhance application security.

Inbound vs Outbound rules

Inbound rules are firewall rules that control what traffic is allowed to enter a resource. I configured an inbound rule that permits HTTP traffic on port 80 and SSH traffic on port 22 to ensure secure access and web functionality.

Outbound rules are firewall rules that control what traffic is allowed to leave a resource. By default, my security group's outbound rule permits all traffic to leave the instance, ensuring unrestricted outbound communication.

The screenshot displays the AWS Management Console interface for a security group. The title bar reads 'sg-0d6b299c7467bd615 - NextWork Security Group'. Below this, the 'Details' tab is active, showing a grid of information: Security group name (NextWork Security Group), Security group ID (sg-0d6b299c7467bd615), Description (A Security Group for the NextWork VPC), VPC ID (vpc-0a4c28603f48a4789), Owner (011528261245), Inbound rules count (1 Permission entry), and Outbound rules count (1 Permission entry). Below the details, there are tabs for 'Inbound rules', 'Outbound rules', and 'Tags'. The 'Inbound rules' tab is selected, showing a search bar and a table with one rule. The table has columns for Name, Security group rule..., IP version, Type, Protocol, and Port range. The rule is listed with a checkbox on the left and a search icon on the right.

Name	Security group rule...	IP version	Type	Protocol	Port range



Network ACLs

Network ACLs are stateless firewalls in a VPC that control inbound and outbound traffic at the subnet level. They allow or deny specific IP ranges, protocols, and ports, providing an additional layer of security beyond security groups.

Security groups vs. network ACLs

The difference between a security group and a network ACL is that security groups are stateful and control traffic at the instance level, while network ACLs are stateless and control traffic at the subnet level, allowing or denying specific traffic.



Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules will allow all traffic in both directions. These default rules permit any IP, protocol, and port to communicate, but they can be modified for more restrictive control.

In contrast, a custom ACL's inbound and outbound rules are automatically set to deny all traffic. You must explicitly create allow rules for any desired communication, making custom ACLs more restrictive by default than the default ACL configuration.

Inbound rules

Outbound rules

Subnet associations

Tags

Inbound rules (2)

Filter inbound rules

Edit inbound rules

< 1 > ⚙

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny



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